

HYDROGEN SULFIDE**ICSC: 0165 (April 2017)**Dihydrogen sulphide
Hydrogen sulphide**CAS #: 7783-06-4****UN #: 1053****EC Number: 231-977-3**

	ACUTE HAZARDS	PREVENTION	FIRE FIGHTING
FIRE & EXPLOSION	Extremely flammable. Gas/air mixtures are explosive.	NO open flames, NO sparks and NO smoking. Closed system, ventilation, explosion-proof electrical equipment and lighting. Prevent build-up of electrostatic charges (e.g., by grounding) if in liquid state. Do NOT use compressed air for filling, discharging, or handling.	Shut off supply; if not possible and no risk to surroundings, let the fire burn itself out. In other cases extinguish with water spray, dry powder. In case of fire: keep cylinder cool by spraying with water.

AVOID ALL CONTACT! IN ALL CASES CONSULT A DOCTOR!

	SYMPTOMS	PREVENTION	FIRST AID
Inhalation	Cough. Headache. Dizziness. Nausea. Laboured breathing. Shortness of breath. Irregular heartbeat. Convulsions. Unconsciousness.	Use ventilation, local exhaust or breathing protection.	Administration of oxygen may be needed. Fresh air, rest. Half-upright position. Artificial respiration may be needed. No mouth-to-mouth artificial respiration. Refer immediately for medical attention.
Skin	ON CONTACT WITH LIQUID: FROSTBITE.	Cold-insulating gloves.	ON FROSTBITE: rinse with plenty of water, do NOT remove clothes. Refer for medical attention .
Eyes	Redness. Pain.	Wear safety goggles or eye protection in combination with breathing protection.	First rinse with plenty of water for several minutes (remove contact lenses if easily possible), then refer for medical attention.
Ingestion		Do not eat, drink, or smoke during work.	

SPILLAGE DISPOSAL	CLASSIFICATION & LABELLING
Remove all ignition sources. Evacuate danger area! Consult an expert! Personal protection: gas-tight chemical protection suit including self-contained breathing apparatus. Do NOT let this chemical enter the environment. Shut off cylinder if possible. Ventilation. Remove gas with fine water spray. Isolate the area until the gas has dispersed.	According to UN GHS Criteria  DANGER Extremely flammable gas Contains gas under pressure; may explode if heated Fatal if inhaled Causes serious eye irritation May cause respiratory irritation Very toxic to aquatic life
STORAGE	
Fireproof. Separated from strong oxidants. Cool. Keep in a well-ventilated room. Install continuous monitoring system with alarm. Provision to contain effluent from fire extinguishing. Store in an area without drain or sewer access.	
PACKAGING	
	Transportation UN Classification UN Hazard Class: 2.3; UN Subsidiary Risks: 2.1



Prepared by an international group of experts on behalf of ILO and WHO, with the financial assistance of the European Commission.
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HYDROGEN SULFIDE**ICSC: 0165****PHYSICAL & CHEMICAL INFORMATION****Physical State; Appearance**

COLOURLESS COMPRESSED LIQUEFIED GAS WITH CHARACTERISTIC ODOUR OF ROTTEN EGGS.

Physical dangers

The gas is heavier than air and may travel along the ground; distant ignition possible. As a result of flow, agitation, etc., electrostatic charges can be generated.

Chemical dangers

Heating may cause violent combustion or explosion. Decomposes on burning. This produces toxic gases of sulfur oxides. Reacts with strong oxidants. This generates fire and explosion hazard. Attacks many metals and some plastics.

Formula: H₂S

Molecular mass: 34.1

Boiling point: -60°C

Melting point: -85°C

Solubility in water, g/100ml at 20°C: 0.5

Relative vapour density (air = 1): 1.19

Flash point: Flammable gas

Auto-ignition temperature: 270°C

Explosive limits, vol% in air: 3.9-45.5

Vapour pressure, kPa at 20°C: 1880

Relative density (water = 1): 0.92

EXPOSURE & HEALTH EFFECTS**Routes of exposure**

The substance can be absorbed into the body by inhalation.

Effects of short-term exposure

Rapid evaporation of the liquid may cause frostbite. The substance is irritating to the eyes and respiratory tract. Inhalation of this gas may cause lung oedema. The effects may be delayed. Medical observation is indicated. See Notes. The substance may cause effects on the central nervous system. Exposure could cause unconsciousness. Exposure could cause death.

Inhalation risk

A harmful concentration of this gas in the air will be reached very quickly on loss of containment.

Effects of long-term or repeated exposure**OCCUPATIONAL EXPOSURE LIMITS**

TLV: 1 ppm as TWA; 5 ppm as STEL.

MAK: 7.1 mg/m³, 5 ppm; peak limitation category: I(2); pregnancy risk group: C.

EU-OEL: 7 mg/m³, 5 ppm as TWA; 14 mg/m³, 10 ppm as STEL

ENVIRONMENT

The substance is very toxic to aquatic organisms.

NOTES

The symptoms of lung oedema often do not become manifest until a few hours have passed and they are aggravated by physical effort.

Rest and medical observation are therefore essential.

Specific treatment is necessary in case of poisoning with this substance; the appropriate means with instructions must be available.

Since the warning effect of the odour may be absent above the OEL (paralysis of the olfactory nerve), there is considerable risk of intoxication. The odour cannot be relied on as an adequate warning of the presence of hydrogen sulfide.

ADDITIONAL INFORMATION**EC Classification**

Symbol: F+, T+, N; R: 12-26-50; S: (1/2)-9-16-36-38-45-61

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